Fluence Calculation

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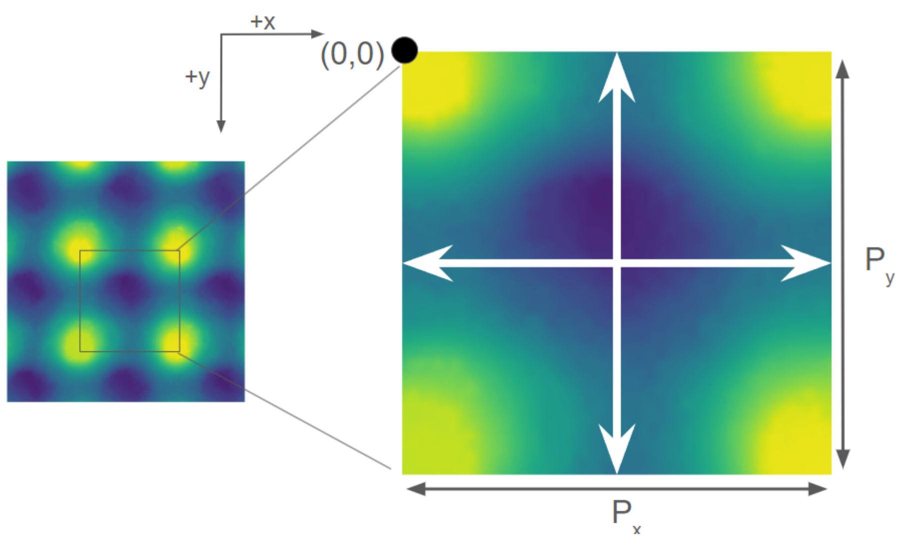
The fluence/intensity at a position (x, y) from a single beam centered at (x_0, y_0) is given by

$$I(x,y) = I_0 e^{-\frac{(x-x_o)^2+(y-y_o)^2}{\sigma^2}} \tag{1}$$

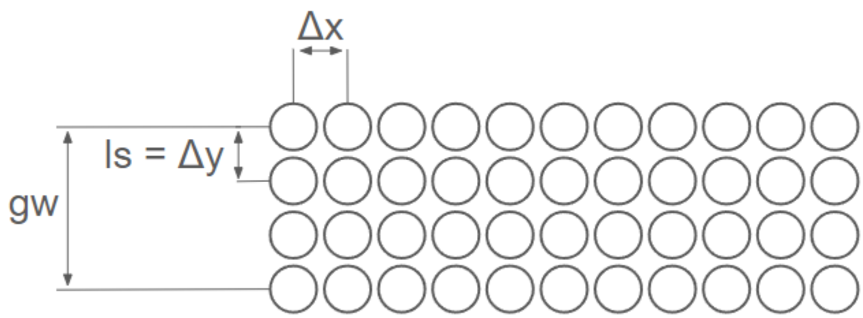
Thus, the fluence of multiple beams is simply their sum $I_{tot} = \sum_{all\ beams} I_{beam}$

Now, consider a "unit cell" with dimensions of $P_x \times P_y$. At each corner of the "unit cell" is



the peak of a pyramid. I will define the origin as the top left corner of this unit cell. There will be two lines passing through this unit cell where passes are made: one horizontal and one vertical.

Let's first consider the horizontal line. The total fluence at a position (x, y) from the beams in this horizontal row is $I_{row} = \sum_i \sum_j I_0 e^{-\frac{(x-x_i)^2+(y-y_i)^2}{\sigma^2}}$. As shown in figure x, x_i will be $i \cdot \Delta x$ and y_i will be $j \cdot \Delta y + \frac{P_y - gw}{2}$, where $\Delta x = \frac{SS}{RR}$ and $\Delta y = ls$. In the calculation, I will include



the contribution from all beams within an area concentric to the unit cell with dimensions $1.1P_x \times 1.1P_y$. Thus, i will range from $\frac{-0.05P_x}{\Delta x}$ to $\frac{1.05P_x}{\Delta x}$. j will range from 0 to $\frac{gw}{ls}$. Plugging all of these values in, the total fluence from the horizontal line is

$$I_{row} = \sum_{i=\frac{-0.05P_x}{\frac{SS}{RR}}}^{\frac{1.05P_x}{\frac{SS}{RR}}} \sum_{j=0}^{\frac{gw}{ls}} I_0 e^{-\frac{(x-\frac{SS}{RR}i)^2+(y-(j\cdot ls+\frac{P_y-gw}{2})^2}{\sigma^2}} \tag{2}$$

The exact same logic applies to the vertical line, only x and y are switched. The result is

$$I_{col} \sum_{i=0}^{\frac{gw}{ls}} \sum_{j=\frac{-0.05P_y}{\frac{SS}{RR}}}^{\frac{1.05P_y}{\frac{SS}{RR}}} I_0 e^{-\frac{(x-(i\cdot ls+\frac{P_x-gw}{2})^2)+(y-j\frac{SS}{RR})^2}{\sigma^2}} \tag{3}$$

So, the total fluence per layer is just $I_{row} + I_{col}$. Multiplying by the number of layers gives $I_{tot}(x, y) =$

$$n\left(\sum_{i=\frac{-0.05P_x}{\frac{SS}{RR}}}^{\frac{1.05P_x}{\frac{SS}{RR}}} \sum_{j=0}^{\frac{gw}{ls}} I_0 e^{-\frac{(x-\frac{SS}{RR}i)^2+(y-(j\cdot ls+\frac{P_y-gw}{2})^2}{\sigma^2}} + \sum_{i=0}^{\frac{gw}{ls}} \sum_{j=\frac{-0.05P_y}{\frac{SS}{RR}}}^{\frac{1.05P_y}{\frac{SS}{RR}}} I_0 e^{-\frac{(x-(i\cdot ls+\frac{P_x-gw}{2})^2)+(y-j\frac{SS}{RR})^2}{\sigma^2}} \right) \tag{4}$$